

Concentrated Fire: The Effects of Defense Industry Consolidation on Government Procurement

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Abstract

This paper studies the impact of mergers between large defense contractors on procurement costs of high-profile Department of Defense (DoD) acquisition programs. Using novel program-level data, I analyse changes in the cost growth of acquisition programs exposed to the consolidation effects of six large-scale mergers during the 2010s. Large horizontal and vertical mergers did not significantly affect procurement costs of major weapons systems. DoD buyer power appears to blunt the markup effects of market concentration, but this effect may be largely limited to the production phase.

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1 Introduction

Annual expenditures on national defense constitute approximately half of the United States government’s discretionary spending. Compared to other corners of government activity, defense spending is unusual in how heavily it relies on private contractors. On an annual basis, contracts with these firms capture approximately half of the DoD’s total budgetary obligations (Sanders, Jang, and Holderness 2022).¹ The industry in which these firms operate is also characterized by a rigid and concentrated market structure. However, the effects of concentration in this space are theoretically ambiguous. On one hand, consolidation may endow prime contractors with more power to charge markups on the systems they supply to the government. On the other hand, the bargaining power wielded by the DoD as a monopsonist may offset the market power gained through a merger in ways that constrain costs.

What then, does the increasing degree of market power wielded by these firms actually mean for the governments whose militaries they furnish with hardware? Throughout the 2010s, several prime contractors participated in a series of large-scale mergers and acquisitions. I take advantage of the resulting increase in market concentration to investigate this question. My analysis draws on novel, hand-collected administrative data on the costs of high-profile, multi-year DoD acquisition programs. I combine this with an event study design that exploits the staggered timing of these mergers to estimate the effects of a contractor’s merger on the costs of acquisition programs in which they play a leading role.

My results indicate that contractor consolidation did not significantly affect procurement costs during this period. High-profile acquisition programs run by non-merging contractors show cost trajectories statistically indistinguishable from those exposed to a merger by their prime contractor. In particular, I estimate post-merger changes in both procurement cost *levels* and cost *growth* that while small, are statistically insignificant. Nonetheless, I believe these results are informative and provide some insight into the mechanics of defense contracting, namely in terms of market structure and incentives.

A substantial empirical literature base connects market structure to market outcomes, a tradition which this paper builds on. Landmark studies include Reiss and Spiller (1989), who analyse the competition effects of entry in small airline markets, as well as Bresnahan and Reiss (1991), who show how competitive conduct changes following entry in concentrated markets. I also contribute to the small rank of studies analysing the defense industry

¹Sanders, Jang, and Holderness (2022) also note that in recent years this figure is actually closer to 60%.

and its attendant contracting environment, a corner of literature which is disproportionately small relative to the share of government activity that defense contracting constitutes. Notable contributions include [Hensel \(2010\)](#), who offers a simple descriptive analysis of mergers and procurement costs, as well as [Bhattacharya \(2021\)](#), who examines how the DoD structures incentives in R&D contests for small contractors and what the optimal design is. But the most relevant study in this regard is [Carril and Duggan \(2020\)](#), who study the effects of a horizontal consolidation wave between prime contractors in the 1990s on patterns in DoD procurement, finding that mergers made contracting processes less competitive, but did not affect procurement costs. These studies provide a solid foundation, but they also predate the most recent round of contractor consolidation. I fill an important gap in this area by extending the analysis to a series of large-scale mergers during the 2010s, which have received almost no empirical attention.

The rest of this paper is structured as follows: [Section 2](#) presents some relevant background information and institutional context. [Section 3](#) provides an overview of the data and details its key features, while [Section 4](#) describes the research design and econometric strategy. [Section 5](#) reports the primary empirical results as well as findings from some supplementary analyses. [Section 6](#) provides a brief discussion and interpretation of the findings. [Section 7](#) concludes by highlighting some of the limitations and proposes some avenues for future research.

2 Background

2.1 DoD Acquisition: Terms and Regulations

Acquisition programs are the primary vehicle through which the Department of Defense procures weapons systems. In addition, they account for a sizeable share of the DoD's annual procurement and R&D expenditures. The size and scope of these programs has thus made them subject to a formal categorization and oversight regime intended to keep the programs accountable to schedule and budget commitments.

Every acquisition program is broadly classified into an acquisition category (ACAT). This classification is dictated by two main factors: the program's expected cost, and the degree of interest in the capabilities being acquired ([DoDI 5000.85 2020](#)). The most high-profile acquisition programs in the DoD's portfolio are designated as ACAT I, typically encompassing major weapon systems or large-scale modernisation programs. Any program in ACAT I is also referred to as a Major Defense Acquisition Program (MDAP), which is for-

mally defined as any program that “is estimated to require an eventual total expenditure for research, development, and test and evaluation of more than \$525 million in Fiscal Year (FY) 2020 constant dollars or, for procurement, of more than \$3.065 billion in FY2020 constant dollars” ([DoDI 5000.85 2020](#)). Below ACAT I, programs are further segmented into ACAT II and III based on decreasing cost and scope.

The degree of oversight and administrative procedures that an acquisition program is governed by is also dictated by the acquisition category. MDAPs in particular are subject to additional transparency and statutory reporting requirements that programs in lower tiers such as ACAT II/III are not. Oversight of MDAPs is administered through Selected Acquisition Reports (SARs), annual filings documenting a program’s cost, schedule, and performance. These reports are updated annually and the Department of Defense is legally obligated to submit them to Congress by the end of each calendar year. SARs are also the vehicle through which one of the system’s sharpest accountability mechanisms is enforced. The Nunn-McCurdy Act requires the DoD to notify Congress if an MDAP experiences growth in unit costs in excess of a specific threshold ([Neenan 2025](#)). Depending on the severity of the breach, the DoD may be obligated to restructure the program to bring it back into compliance. Exceptionally serious breaches may also result in a program being terminated outright. This particular combination of statutory reporting and cost growth accountability makes MDAPs the setting for the analysis that follows.

2.2 Period of Study: Industry Consolidation

The paper’s analysis spans the fourteen-year window from 2010 to 2023. I choose to focus on this period for two reasons. First, much like the consolidation wave of the 1990s, market concentration in the defense industry underwent a similarly sharp (if not more acute) increase during the 2010s. Second, merger deals during this period were characterized by a greater degree of heterogeneity compared to the purely horizontal plays of 1990s mergers.

After a period of comparatively little M&A activity during the 2000s, the 2010s saw another wave of major mergers among DoD prime contractors. In 2019, Raytheon and United Technologies announced a merger agreement valued at \$121 billion, by far the largest M&A transaction ever recorded in the defense industry ([Fontanella-Khan and Pfeiffer 2019](#)). This closely followed other significant deals, such as the 2018 merger between Rockwell Collins and United Technologies, as well as the 2015 acquisition of Sikorsky Aircraft by Lockheed Martin. Similar to the consolidation wave two decades prior, many of the contractors involved in these deals were already among the largest at the start of the decade. The end

result was an increasing share of contract dollars accruing to an ever-smaller set of firms. By 2020, contract obligations awarded to the “big five” prime contractors (Lockheed Martin, Raytheon, Boeing, Northrop Grumman, and General Dynamics) reached 36%, their highest share in over two decades (Sanders, Jang, and Holderness 2022). A summary of the six largest mergers in this period is presented in Table 1.

Table 1: Major Mergers Among DoD Contractors (2010 – 2020)

Merging Contractors	Year	Product Market	Merger Type
Goodrich – United Technologies	2012	Aircraft	Product-Extension
Lockheed Martin – Sikorsky Aircraft	2015	Aircraft (Rotary-Wing)	Product-Extension
Northrop Grumman – Orbital ATK	2018	Ordnance, Space Systems	Vertical
Rockwell Collins – United Technologies	2018	Aircraft	Product-Extension
L3 Technologies – Harris Corporation	2019	Electronics	Horizontal
Raytheon – United Technologies	2020	Aircraft, Electronics, Ordnance	Conglomerate

Note: I identify the six listed mergers using data from Refinitiv Workspace. I filter merger transactions to those where: (i) the closing date is between 2012 and 2020, (ii) either firm is categorised in the “Aerospace and Defense” industry, and (iii) both firms were among the top 30 defense contractors in 2010 by arms revenues.

Unlike the 1990s wave, this round of consolidation was not primarily horizontal. Deals during this period were largely defined by an emphasis on vertical integration and supply chain consolidation. This is hardly incidental, and can be viewed as a logical consequence for the industry following the 1990s consolidation wave. By 1998, the frenzy of horizontal mergers had collapsed the number of aerospace and defense prime contractors from fifty-one to just five (OUSD 2022). At this point, there remained almost no opportunities for further horizontal consolidation between prime contractors that would survive antitrust review.² Having completely exhausted their horizontal options, contractors seeking to consolidate were effectively forced to shift toward vertical acquisitions or product extension mergers.³ The dominant theme of deals during this period thus became one of plays to consolidate supply chains, either via acquiring key suppliers or by forming conglomerates to extend market footprint. The resulting variation in deal structure potentially carries different implications for market outcomes, particularly since vertical, horizontal, and conglomerate mergers confer market power in different ways. This distinction is central to the empirical analysis outlined in Section 4.

²The most notable example was the proposed 1998 mega-merger between Lockheed Martin and Northrop Grumman, which was abandoned after regulators sued to block the deal (Carril and Duggan 2020).

³Not all recent vertical deals were able to escape antitrust scrutiny. In 2022, Lockheed Martin terminated its acquisition of Aerojet Rocketdyne, a major manufacturer of rocket propulsion systems after the FTC sued to block the deal (FTC 2022).

3 Data

For data on acquisition costs, I take advantage of the fact that the DoD is required to prepare annual reports of acquisition programs for transparency and accountability purposes. As discussed earlier, the standardized format and long time-horizon of these reports make them ideal for tracking acquisition outcomes. Each SAR runs dozens of pages, documenting schedule, funding breakdowns, and other parameters in granular detail. However, relevant cost data is reported in a single page containing tables with estimates of program costs on an aggregate and per-unit basis. Each table provides the original baseline estimates available at a program's inception, as well as the most recent end-of-year cost estimates. Figure 1 shows an example of a typical SAR and its attendant unit cost report.

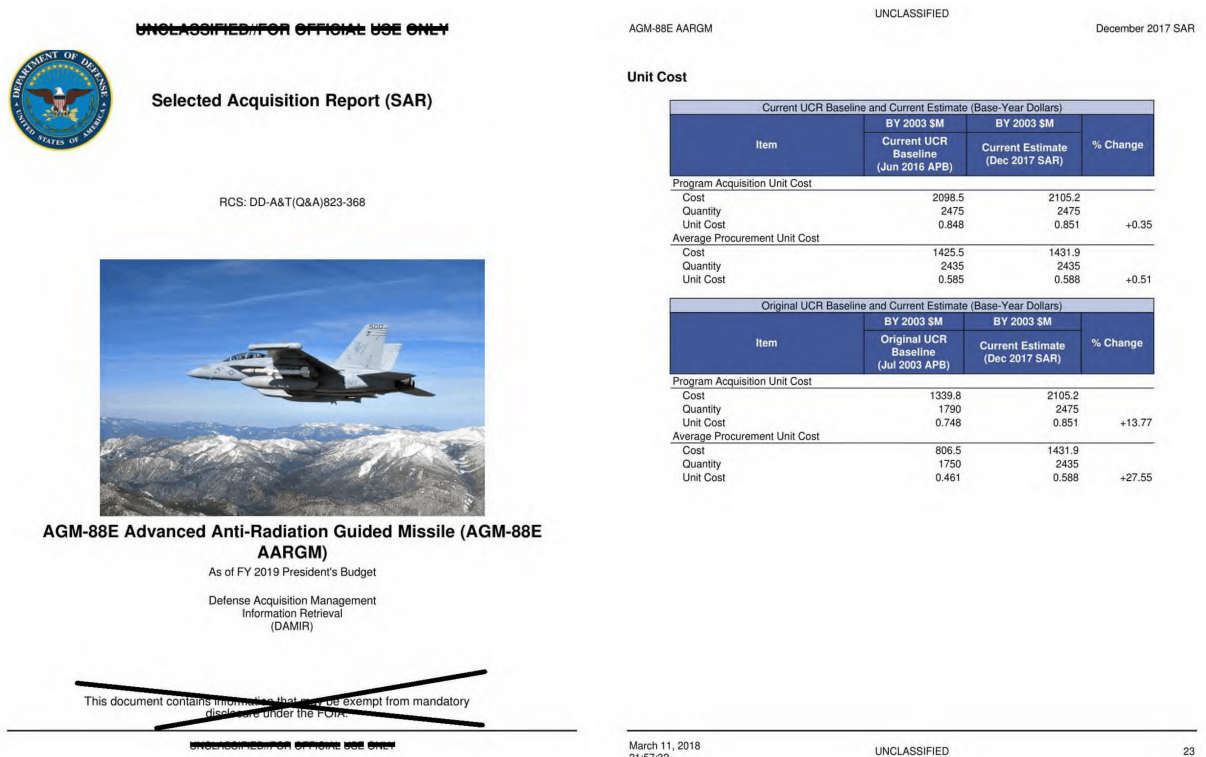


Figure 1: Example SAR and Unit Cost Report

These unit cost reports present two different estimates of program costs. The first is total *procurement* cost, which includes all recurring and non-recurring costs associated with the production of a weapon system. Average procurement unit cost (APUC) averages this figure over procurement quantity, and represents the per-unit cost to the DoD solely of purchasing the end item. The second measure reported is total *acquisition* cost, which includes all program related expenditures on top of procurement, such as R&D and acquisition-phase

maintenance. This is also reported in the tables on a per-unit basis as program acquisition unit cost (PAUC). An advantage of these cost reports is that all estimates are reported in constant dollars of the program's original base year, thereby isolating changes in costs attributed to revisions in scope, contract modifications, delays, and so on. This makes them useful for tracking changes in program costs over time.

Drawing on unit cost reports in available SARs from 2010 to 2023, I compiled this information into an unbalanced panel dataset which tracks program cost estimates and their respective baselines for all MDAPs active during this period. The resulting dataset has 1,151 program (*i*) – year (*t*) observations across 150 unique acquisition programs.⁴ Each program-year observation includes information on the current estimated costs and quantities in addition to the original cost and quantity estimates from the base year.⁵ I also observe information on the service branch overseeing the program, the program's original base year, and whether the program is in development or production in any given year. [Table 2](#) presents some descriptive statistics of this dataset.

Another useful piece of information available in SARs is a section which details all active contracts for a particular MDAP. Information documented includes the contract awardee as well as contract type. Using this information, I identify each MDAP's prime contractor. If a SAR lists active contracts with multiple firms and it is not clear who the prime contractor is, I cross-reference this with information from other DoD documents or corporate disclosures to confirm the prime contractor's identity. For a subset of programs, secondary contractors are also identified if their role is both significant to the program's primary cost structure, such as propulsion, engine systems, or joint ventures and is also independently documented in other government acquisition documents or corporate disclosures.

Finally, I map each observed acquisition program to a product market. I define product markets using “platform portfolios”, a classification system used by the Center for Strategic and International Studies to categorize DoD contract obligations and other related expenditures ([Sanders, Jang, and Holderness 2022](#)). Under this definition, each MDAP is assigned into one of seven categories: (1) aircraft, (2) air and missile defense, (3) electronics, communications, and sensors, (4) land vehicles, (5) ordnance and missiles, (6) ships and sub-

⁴For analysis, I exclude three outlier programs widely viewed as structural procurement failures by the DoD. In these cases, large slashes in procurement quantities prior to program cancellation resulted in unit cost growth of several hundred percent, making them unrepresentative of acquisition dynamics.

⁵Several programs had their base years updated due to changes in program milestones. For consistency, I rebase all subsequent estimates for these programs to their original base year using deflators from Table 5-9 of the *National Defense Budget Estimates for FY2024*, also known as the DoD “Green Book”.

marines, and (7) space systems. This taxonomy forms the basis for the empirical strategy that I turn my attention to in the next section.

4 Empirical Strategy

4.1 Research Design

To assess the effect of contractor consolidation on acquisition costs, I exploit the staggered timing of mergers between large defense contractors during this period using a dynamic event study design. The primary design challenge is that unlike horizontal mergers, vertical and product extension mergers confer market power in overlapping product spaces rather than across an entire market.⁶ Defining a program as “treated” solely on the basis of contractor involvement is thus an insufficient criteria for a few reasons. For instance, it risks biasing or attenuating consolidation effects by including markets with limited exposure to specific mergers. In addition, mergers of this type are often scrutinized on a product-line basis by antitrust authorities assessing possible competitive implications (see [Section 6](#) for a more detailed discussion).

As such, I follow [Carril and Duggan \(2020\)](#) and define an MDAP as treated (i.e. exposed to a merger) if the program’s prime contractor or major subcontractor is involved in one of the mergers listed in [Table 1](#), adding the condition that the MDAP’s product market also overlaps with one of the product markets spanned by the merger.⁷ This builds on the approach of previous studies, while also adapting the treatment in order to isolate channels a merger could plausibly operate through.

To illustrate how treatment assignment works, consider the case of Lockheed Martin’s merger with Sikorsky, a major manufacturer of rotary-wing aircraft. Under this merger, all of Sikorsky’s helicopter programs are defined as treated, as are any helicopter programs for which Lockheed is a major subcontractor or systems integrator (e.g. Apache).⁸ On the other hand, programs in Lockheed’s other portfolios such as the PAC-3 air defense interceptors or the JASSM-ER standoff weapon remain untreated under this heuristic. While this is a

⁶Raytheon–United Technologies is a slight exception. Given the scale of the deal and the resulting conglomerate’s substantial market footprint, I extend the overlap condition across its three major product portfolios.

⁷Contractor involvement also includes programs where the prime contractor is a subsidiary of a firm appearing in one of the listed mergers. An example is Pratt & Whitney, a major manufacturer of jet engines which is a business unit of United Technologies (now RTX Corporation).

⁸For the Apache program in particular, Lockheed contributes fire control radars via Longbow LLC, a joint venture with Northrop Grumman ([Lockheed Martin 2026](#)).

slightly more restrictive treatment rule, it more accurately reflects market segmentation in the defense industry. Acquiring Sikorsky is unlikely to have increased Lockheed’s market power in the guided missiles and munitions space.

To further illustrate the treatment heuristic, consider also the acquisition of Orbital ATK by Northrop Grumman. For many years, Orbital ATK was a major supplier of solid rocket motors and a prime contractor on tactical missile programs for the U.S. Navy. Under this merger, treated units include: (i) any ordnance program run by Orbital ATK, such as the AGM-88E anti-radar missile; and (ii) any program with Orbital ATK as a propulsion supplier, such as the Trident D5 fleet ballistic missile, for which the company supplied all three rocket motor stages ([Orbital ATK, Inc. 2018](#)). Northrop’s aviation and electronics programs, by contrast, remain untreated as they straddle a market unaffected by this particular merger.

4.2 Estimation

In this section, I use standard panel methods to estimate program-level treatment effects. My primary event study specification is of the form:

$$y_{it} = \alpha_i + \sum_{k \in K} \beta^k \mathbf{1}[k_{it} = k] + \gamma_t + \mathbf{X}_{it}' \boldsymbol{\Omega} + \nu_{it} \quad (1)$$

where y_{it} represents some measure of program cost for program i in year t , α_i are program fixed effects, γ_t are time fixed effects, $\mathbf{X}_{it}$ is a vector of program specific control variables, such as program duration and phase, and ν_{it} is the regression residual. I am interested in the coefficients β^k , which if properly estimated, should capture the causal effect of a merger on acquisition costs at time k . My identification strategy rests on the assumption that absent any merger, the procurement costs of programs exposed to a merger by their prime contractor (e.g. Raytheon) would have followed the same trajectory as programs run by non-merging contractors (e.g. Boeing).

As shown in [Goodman-Bacon \(2021\)](#), a significant concern in settings with staggered treatment rollout such as this one is the possibility that classic TWFE models may produce biased estimates of treatment effects if those effects happen to be time-varying. Given the differential timing as well as the uneven distribution of the six merger cohorts across my sample window, negative weighting poses a potential concern. I therefore estimate Equation (1) using the imputation-based event study estimator proposed by [Borusyak, Jaravel, and Spiess \(2024\)](#). To account for differences in the relative sizes of acquisition programs, I

follow Carril and Duggan (2020) and weight each program-year observation by the natural log of the program’s baseline total acquisition cost estimate. I also cluster standard errors at the program level to account for possible serial correlation within programs.

I consider three different measures of program costs as outcomes. First, I use the total estimated procurement cost of program i in year t (in logs). Second, I use quantity adjusted cumulative growth in APUC relative to its original baseline estimate. Third, I construct a flow variable which measures year-over-year percentage changes in APUC. For measures of unit costs, I opt for APUC instead of PAUC to focus on the cost component associated with production, where I hypothesize market structure effects are likely to be more relevant.

5 Results

5.1 Effects on Acquisition Costs

Figure 2 presents results from the estimation of (1) by imputation for the three outcome variables outlined in Section 4. I also estimate the same equation using a static TWFE model with lags and leads as a de-facto robustness check. Here, the set of control variables includes a quadratic polynomial for program duration, defined as the difference between the current year and a program’s original base year, as well as a binary indicator for whether a program is in production or development in the current year. While production is typically carried out under fixed-price type contracts, research and development stages tend to heavily favor cost-plus contracts. Both of these differ substantially in terms of the financial risk shouldered by the contractor. Absent information on contractual form, the production/development indicator serves as the best available proxy.

From the top panel of Figure 2 we observe that logged procurement costs follow a relatively flat trajectory centered around zero during the pre-merger period. These effects largely continue their null pre-merger trajectory, after which there is a slight downward trend in logged procurement costs beginning four years post-merger. In years five and six, I estimate effects on the order of 8 to 10%, pointing towards some modest decreases in total procurement costs over longer time horizons. However, these effects are statistically indistinguishable from zero.

While aggregate procurement costs exhibit little difference following a merger, a more representative measure of acquisition dynamics is the change in *unit costs*. Since programs frequently encounter revisions in procurement quantities, changes in total procurement

costs driven by changes in quantities ordered by the DoD may mask the possibility that the underlying unit costs remain stable. The middle panel presents event study estimates for year-over-year changes in procurement unit costs. Estimates for treated programs cluster within ± 0.03 of zero across the entire post-merger period, indicating no substantial change in annual unit cost growth following a merger by their prime contractor or major subcontractor. This is further corroborated in the bottom panel, where the outcome is cumulative growth in unit costs. Similar to year-over-year changes in procurement unit costs, we observe estimated effects continuing their flat pre-merger trend and remaining tightly clustered around zero across the entire post-merger horizon. Cost growth trajectories do not seem to be significantly affected by contractor consolidation.

I also investigate whether contractor consolidation affected procurement costs for programs *within* prime contractors who merged. As discussed earlier in [Section 4](#), a merger or acquisition does not automatically confer market power across every product space a firm operates in. Large defense contractors in particular frequently silo different product lines into separate corporate divisions or business units. Lockheed Martin’s corporate structure for instance is comprised of separate operating divisions for aeronautics, missiles and fire control, rotary systems, and so on. I therefore restrict attention to a subsample of programs with prime contractors or major subcontractors appearing in one of the mergers in [Table 1](#).

[Figure 3](#) presents event study estimates for the estimation of (1) on this subsample. As above, I use the same set of control variables for program duration and phase. Compared to the full-sample specification, logged procurement costs appear to temporarily increase following a merger before exhibiting a similar (though slightly less pronounced) dip in years five and six. In any case, these estimates are not reliably distinguishable from zero. Annual and cumulative growth in procurement unit costs show no discernible post-merger change either. Even within merging contractors, there is little evidence to suggest that programs exposed to a merger display meaningfully different changes in procurement costs.

5.2 Supplementary Analyses

5.2.1 Vertical Foreclosure

Are the above results significantly affected by vertical foreclosure? As the sole purely vertical merger during this period, Northrop Grumman’s acquisition of Orbital ATK poses some possible competitive implications.⁹ This deal gave a leading prime contractor potential

⁹See [Rey and Tirole \(2007\)](#) for a detailed treatment of vertical foreclosure.

captive access to the DoD's largest supplier of solid rocket motors as well as a premier manufacturer of tactical missiles and warheads. To investigate this question, I use the previous subsample of acquisition programs run by merging contractors and exclude the six programs in the Northrop-Orbital merger cohort. I then re-estimate (1) on this new subsample to assess how much unit cost trajectories change in the absence of this vertical merger.

Event study estimates for this sample are available in [Figure 4](#). From this, I conclude that the Northrop-Orbital ATK merger did not significantly distort MDAP procurement costs. Assuming that this merger led to vertical foreclosure effects, one would expect to see slightly smaller post-merger cost effects in the absence of programs in this cohort, especially if Northrop Grumman began supplying inputs to rival contractors at a higher markup. This does not seem to be the case. Logged procurement costs follow a post-merger trajectory similar to the previous sample with these programs included. The only noticeable difference is a marginally shorter post-merger period of elevated procurement costs. In any case, the magnitudes of these estimates are also statistically insignificant. There are no noticeable changes in post-merger unit cost growth either with this cohort removed from the sample. I discuss some possible explanations for this in [Section 6](#).

5.2.2 Service Branch Heterogeneity

Does the DoD's monopsony bargaining power apply uniformly across its service branches? In this section, I look at whether contractor consolidation affected cost growth differently across three constituent service branches: Air Force, Army, and Navy.¹⁰ Although the DoD has a tendency to be viewed as a monolith, administration of acquisition programs is largely delegated to acquisition offices at the service branch level. In addition, each of these branches has its own unique organizational culture, which may influence how it conducts procurement.¹¹

[Figures 5-7](#) present event study results for subsets of MDAPs run by each respective service branch. For brevity, I restrict attention to cumulative unit cost growth as the outcome as it is the most informative of the three measures. Army programs exhibit a very slight upward trend in cost growth post-merger, though this is likely a continuation of the pre-merger trend. Again, these effects are not distinguishable from zero. Likewise, cost growth of Navy programs shows no deviation from its null pre-merger trend, and remains tightly

¹⁰While there are some multi-service programs administered by the DoD such as the F-35, this sample is unfortunately too small to reliably draw inferences from.

¹¹See [Ryseff, Welch, and Schneider \(2023\)](#) for a discussion on DoD organizational culture.

clustered around zero. Estimates for Air Force programs are less informative however, and are too noisy to get a read on what the true effects are. This is likely a product of the thin sample of Air Force programs in a merger-exposure cohort. Given the large acquisition footprint of the Air Force, I do not take this analysis as a definitive test.

6 Discussion

Why might have acquisition costs hardly budged despite the sharp increase in market concentration? In this section, I contextualize these results and briefly discuss some of the mechanisms behind them.

6.1 DoD Buyer Power

Market structure is perhaps the most relevant factor which can offset markup pressures following a merger. [Kovacic and Smallwood \(1994\)](#) highlight how the DoD holds a unique dual position, acting as the primary (and sometimes sole) customer for its contractors, while simultaneously regulating how those contractors conduct business. This grants the department a substantial degree of buyer power when negotiating purchases.¹² This is reinforced by the fact that many of these product markets still remain somewhat competitive, thus offering the DoD a menu of credible alternative options for many of the systems or capabilities it may wish to acquire. Even in the face of large-scale consolidation then, the DoD still retains enough power to curb any attempts at charging markups.

6.2 Antitrust Remedies

Buyer power is not the only channel through which the government can keep a lid on procurement costs. Antitrust regulation also confers the government some leverage in this regard. Rather than being rubber-stamped, some of the mergers during this period were subject to behavioral remedies by antitrust authorities as a prerequisite for approval, such as mandatory divestitures or binding clauses designed to mitigate market foreclosure. For instance, Northrop Grumman was required to sign an agreement guaranteeing that it continue supplying Orbital ATK's solid rocket motors to competing prime contractors on a non-discriminatory basis, a measure aimed at preempting potential input foreclosure ([FTC](#)

¹²[Kovacic and Smallwood \(1994\)](#) also caution against treating the DoD as a unitary decisionmaker, noting that acquisition authority is in practice delegated across the military services and individual program offices. Although the branch-level analysis in [Section 5](#) does not point towards systematically different procurement outcomes across services *per se*, the narrow within-branch samples also limit comparisons here.

2018). As a prerequisite for the merger between Raytheon and United Technologies, the companies were required by the DOJ to divest Raytheon’s airborne military radios division as well as United Technologies’ military GPS division to prevent near-total monopolies in these markets (Bartz 2020). In this case, these particular remedies can function as *ex ante* constraints on the exercise of post-merger market power, thereby curbing its pass-through to procurement costs.

6.3 Contracting Incentives

What else can the absence of increased procurement costs be attributed to? Market structure aside, there are additional aspects which possibly play a contributing role. One of the forces long recognized as a source of economic profits for defense firms is the contracting process itself. Rogerson (1994) specifically discusses how fixed-price reimbursement creates a “regulatory lag” that allows contractors to earn profits:

A contract for a single annual lot of missiles or aircraft may easily take two to three years to complete. Furthermore, contracts are often signed six months or more prior to beginning production. The result is that a defense firm will typically have a signed backlog of business equal to two to three times its annual revenue. When a defense firm discovers a way to lower its production costs, previously negotiated prices are not changed. (Rogerson 1994, 72)

Under the incentive structure created by fixed-price contracts, contractors therefore need not resort to charging markups to increase profits on the systems they produce. This is particularly relevant in cases where consolidation acts as a double-edged sword. True, Lockheed Martin’s acquisition of Sikorsky, for instance, grants it significantly more power to sell the end products at a markup. At the same time, efficiency gains from streamlined supply chains could allow the firm to realize the same profits via lower overheads, thereby obviating the need for markups. Given the DoD’s preference for fixed-price contracts at the procurement stage, this dynamic may partially explain why procurement costs remain largely unaffected following a merger.

An ancillary explanation also hinges on the presence of reputation effects in the contracting process. Kovacic and Smallwood (1994) note that the DoD’s repeated dealings with the same contractors allows the department to curb opportunism by making future awards conditional on the state of a contractor’s reputation. Since prime contractors derive a significant share of their revenues from long-term contracts with the DoD, they likely

place a premium on future contract awards. Carril and Duggan (2020) advance a similar argument, noting that “contractors would find it optimal to consider not only the profits accruing from their current portfolio of contracts, but also the expected future contracts that they may obtain from the government.” A contractor which develops a reputation for aggressive markups or cost overruns could plausibly lose out on additional contract opportunities. The promise of future contract revenue may thus act as a disciplinary mechanism which incentivizes contractors to keep costs under control.

7 Conclusion

This paper analyses whether mergers between large defense contractors during the 2010s affected the U.S. government’s procurement costs. Using novel administrative data on DoD acquisition costs, I show that contractor consolidation did not substantially affect the cost trajectories of high-profile acquisition programs. I argue that the DoD’s buyer power and the incentive structure of the contracting process together were able to offset the market power gained by these firms. These findings are broadly consistent with existing studies which find that previous waves of contractor consolidation also had insignificant effects on the DoD’s procurement costs.

The results of this study should be interpreted with some caution. My dataset is strictly limited to large-scale, MDAP-level programs. Lower-tier acquisition programs which also constitute a sizeable share of the DoD’s acquisition portfolio do not enjoy the same level of public-facing transparency. It is possible that these smaller-scale programs have different cost dynamics and that market structure effects operate differently at this level. As such, these results cannot be generalized to defense acquisition as a whole. In addition, the cost estimates I use as outcomes only capture acquisition-phase costs, not the attendant sustainment costs during a system’s entire service life. A firm that forgoes markups to win contracts may have more leverage to extract rents once the government is committed to it as the sole supplier of spares, maintenance, and upgrades. Null results during the acquisition phase do not necessarily rule out consolidation effects that could surface once the contractor’s role shifts to one of sustainment. I acknowledge this as another limitation of my setting.

Nonetheless, I believe my results are informative and shed some light on the mechanics of defense contracting. They highlight how even amidst intense consolidation, the unique structure of this market and the incentives involved can blunt the effects of concentration in

areas where they matter most. Future research might examine whether defense ministries in allied countries such as the UK MoD or Polish MON, which frequently contract with these firms, respond similarly to market concentration. Such work should clarify whether these results are reflective of government bargaining power in general, or just artifacts of U.S. institutions specifically. Examining whether long-term sustainment costs exhibit the same patterns in response to industry consolidation can also clarify whether the government's monopsony power has its limits, and if so, what those limits are.

At the same time, the stakes of this question have changed. With rearmament on the rise across many NATO states, concerns about industry consolidation may take a backseat as industrial capacity becomes a more urgent priority. Future work will hopefully build on this foundation. What was once a question viewed through line items in budget justifications may soon be a question of readiness. *Si vis pacem, para bellum.*

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Appendix

Table 2: Summary Statistics of Acquisition Reports Data

Panel A: Summary Statistics	Mean	Std. Dev.	Min	Max
<i>Program-level variables</i>				
Baseline Acquisition Cost (\$M)	20,051	43,444	779	369,826
Baseline Procurement Cost (\$M)	15,986	35,801	0	300,072
Panel B: Treated vs. Control Programs	All Programs	Treated	Control	
<i>Program-year level variables</i>				
Current Acquisition Cost (\$M)	23,652	17,488	26,140	
Current Procurement Cost (\$M)	18,827	12,645	21,326	
PAUC Growth (%)	1.62%	3.22%	0.96%	
APUC Growth (%)	-0.80%	0.35%	-1.26%	
<i>Platform Portfolio (% of Programs)</i>				
Air and Missile Defense	4.7%	7.1%	4.0%	
Aircraft	28.0%	32.1%	26.6%	
Electronics, Comms, and Sensors	23.3%	28.6%	22.6%	
Land Vehicles	4.0%	0.0%	4.8%	
Ordnance and Missiles	18.7%	21.4%	17.7%	
Ships and Submarines	10.7%	7.1%	12.1%	
Space Systems	10.7%	3.6%	12.1%	
No. Programs	147	28	119	
No. Observations	1,143	327	816	

Notes: All figures are rebased to constant FY2024 dollars using DoD deflators from Table 5-9 of the *National Defense Budget Estimates for FY2024*. Acquisition cost is the total program cost inclusive of R&D expenses. Procurement cost represents the final cost of purchasing the system(s), also defined as acquisition cost less R&D expenses. PAUC/APUC Growth is the mean cumulative percentage change from baseline to current estimates. Treated programs are those assigned to a merger-exposure cohort; control programs are never-treated.

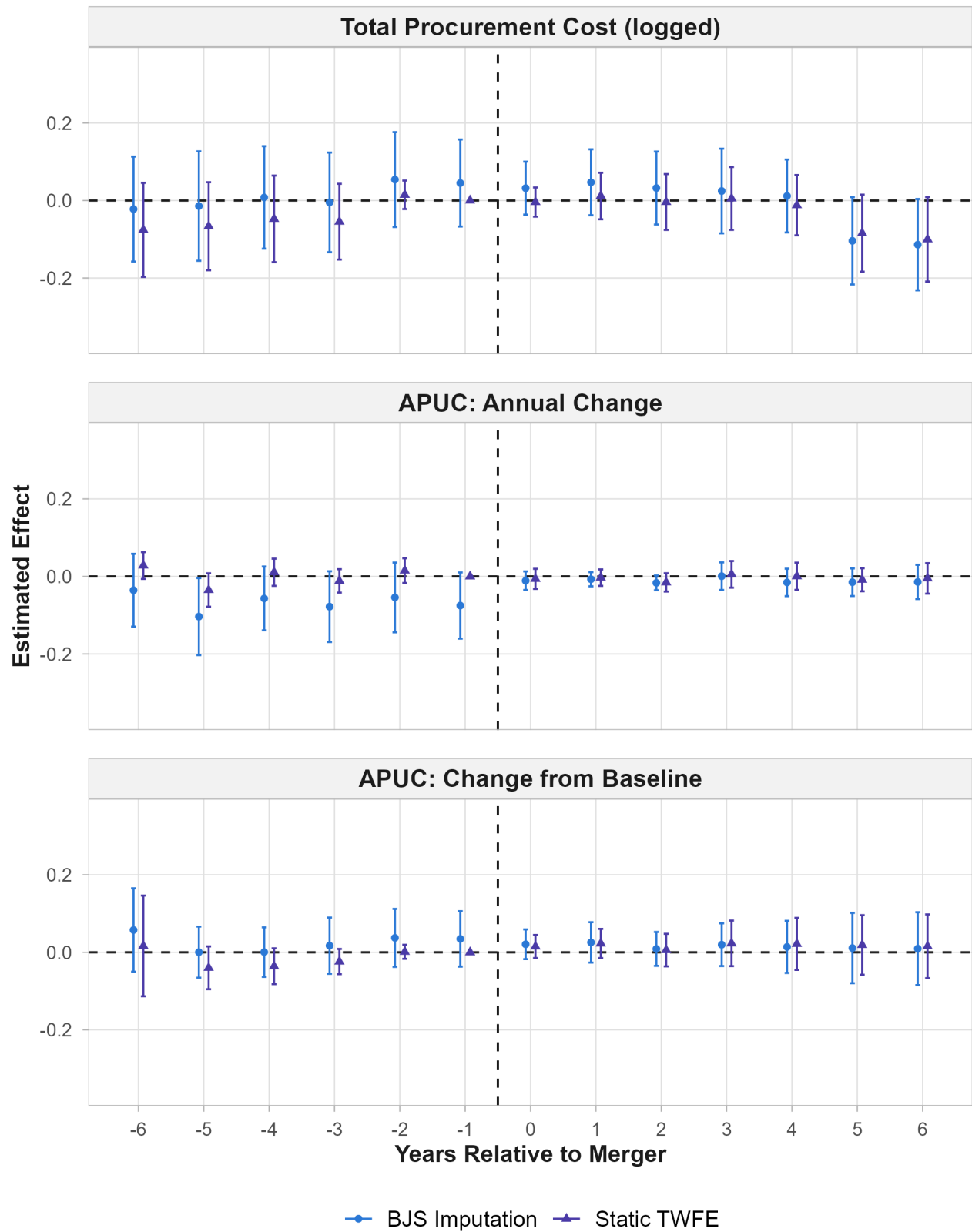


Figure 2: Effect of Mergers on Acquisition Costs (Baseline Sample)

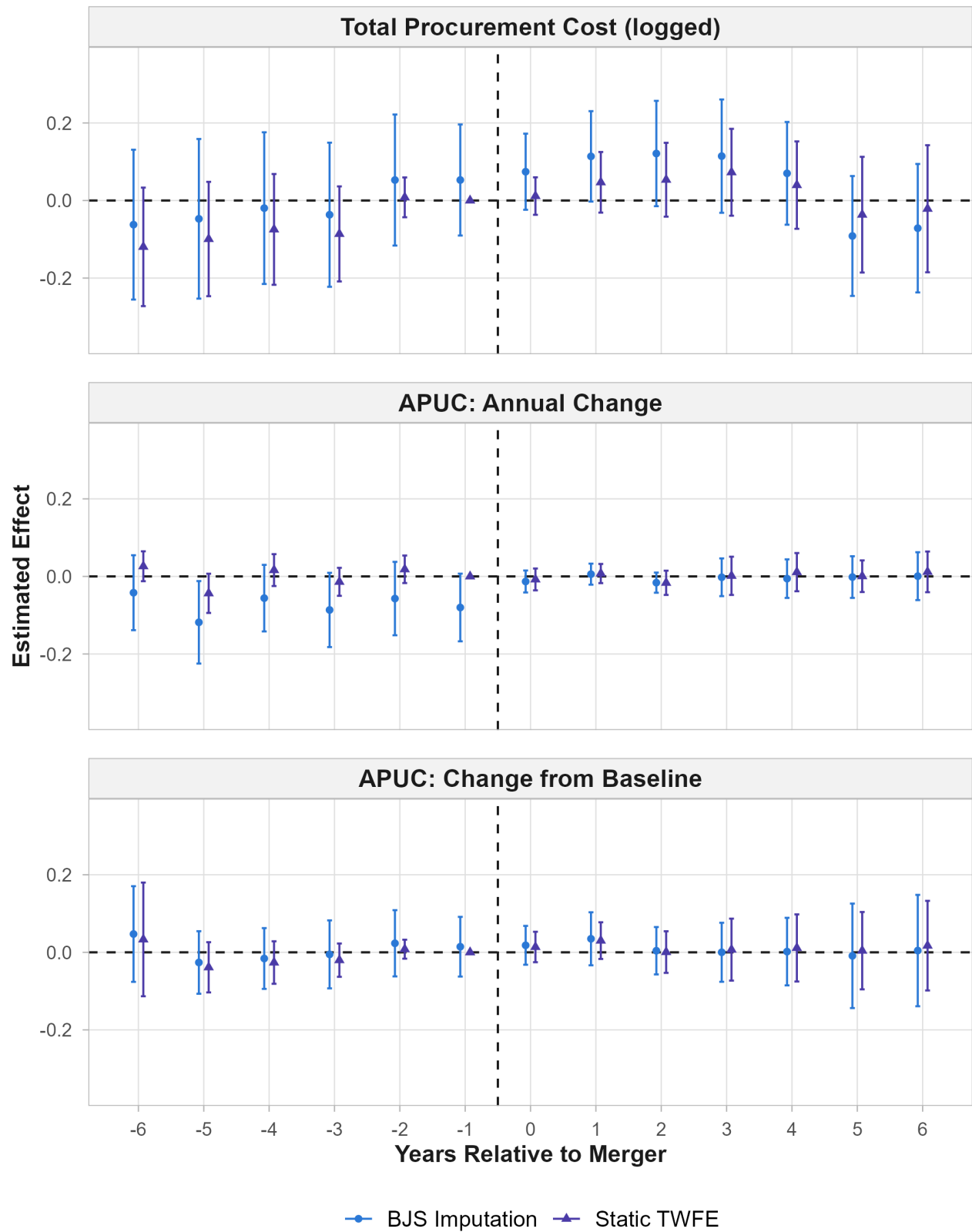


Figure 3: Effect of Mergers on Acquisition Costs (Merging Contractors Only)

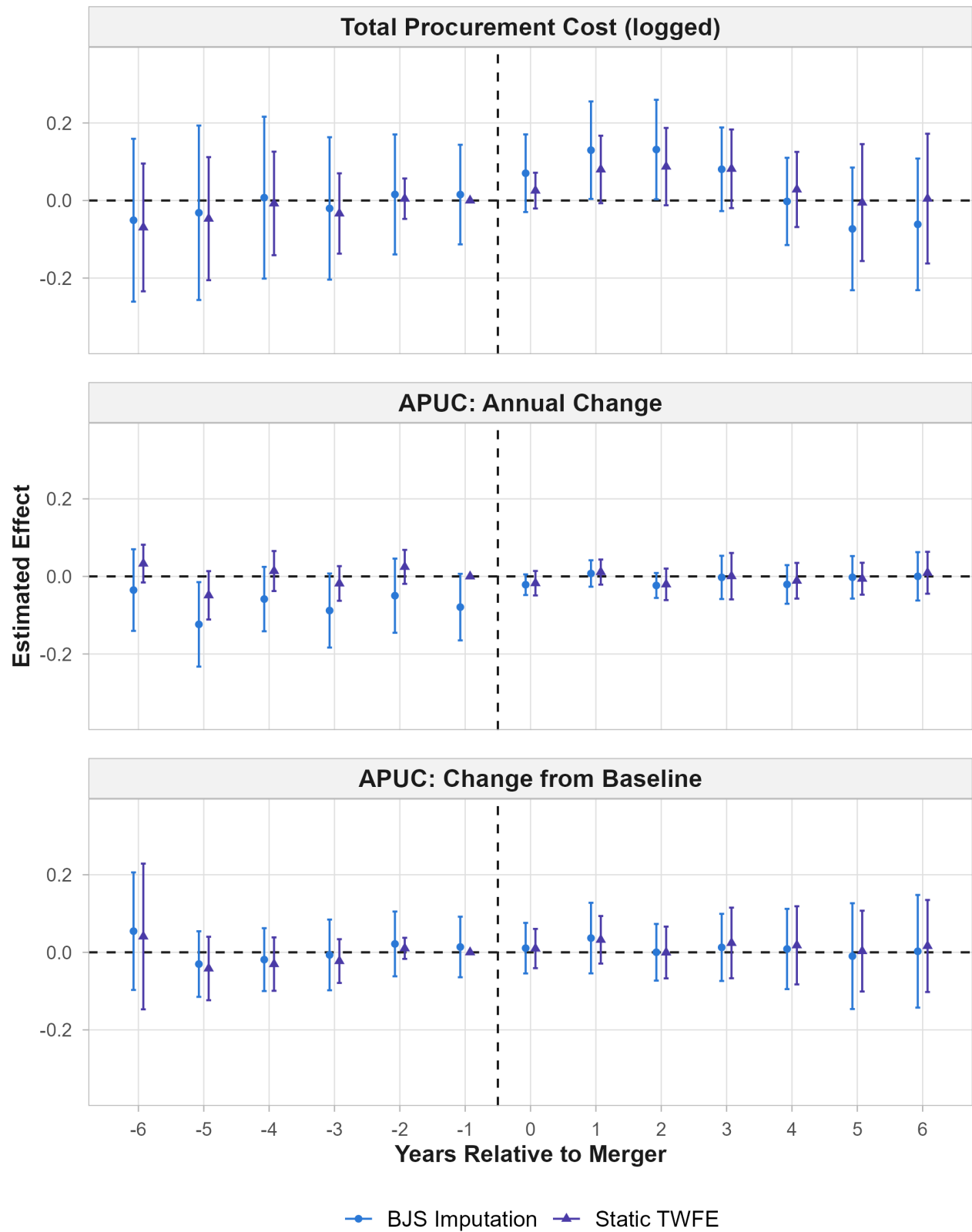


Figure 4: Effect of Mergers on Acquisition Costs (No Orbital ATK Programs)

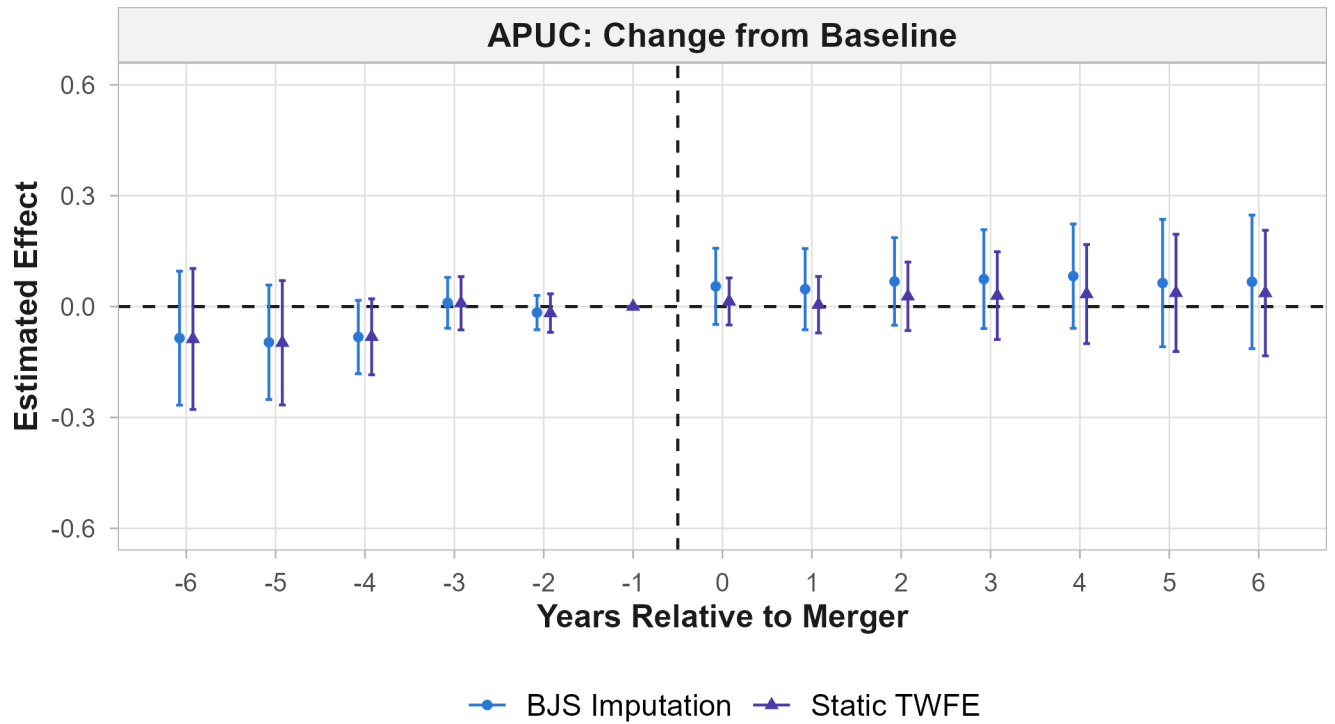


Figure 5: Effect of Mergers on Cumulative Unit Cost Growth (Army Programs)

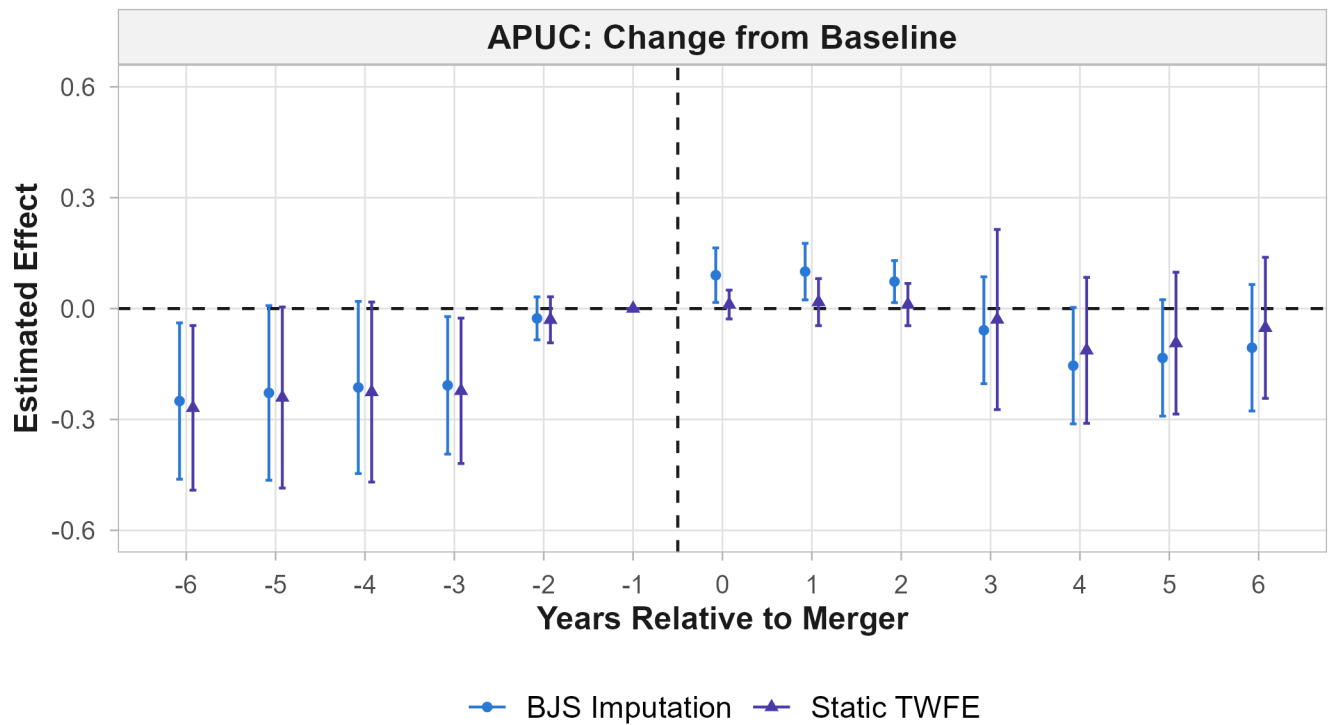


Figure 6: Effect of Mergers on Cumulative Unit Cost Growth (Air Force Programs)

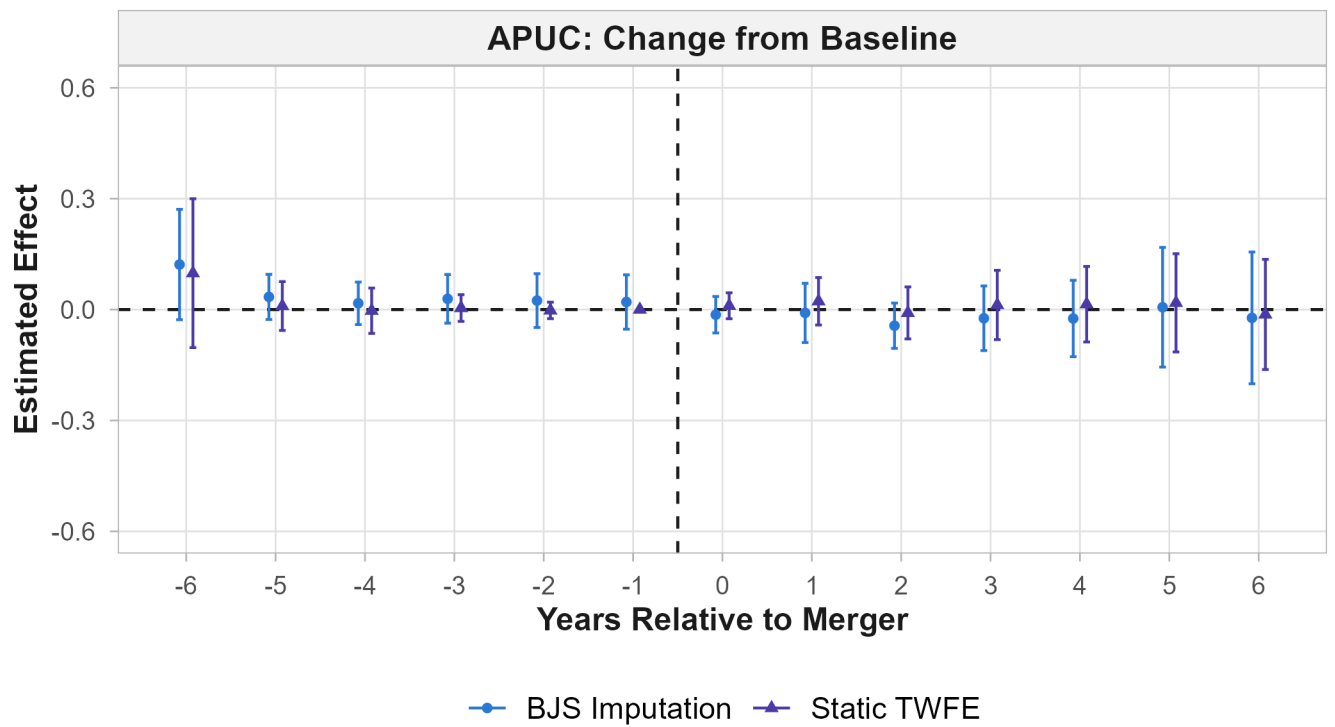


Figure 7: Effect of Mergers on Cumulative Unit Cost Growth (Navy Programs)